

TruthScan

AI-Powered Fake News Detector

Project Documentation

Submitted By:

Darapaneni Vyshnavi

B.Tech — Computer Science Engineering (Data Science)

Dhanekula Institute of Engineering & Technology

Vijayawada, Andhra Pradesh

Version 2.4 | 2025-2026

1. Abstract

TruthScan is a professional, browser-based fake news detection application that uses Natural Language Processing (NLP) heuristics combined with Claude AI deep analysis to classify news articles as credible or disputed. The system analyzes linguistic patterns including sensational language, emotive punctuation, evidence-based writing, and attribution signals to produce a confidence-scored verdict with full explainability. The application operates entirely client-side, ensuring zero data retention and complete user privacy, while delivering results in under one second.

2. Problem Statement

The rapid spread of misinformation and fake news on digital platforms has emerged as one of the most critical challenges of the information age. Research from MIT (Vosoughi et al., 2018) shows that false news spreads six times faster than true news on social media platforms. Key challenges include:

- Over 3.8 billion social media users are exposed to misinformation daily
- 76% of people cannot consistently identify fake news from credible sources
- Manual fact-checking by organizations like Snopes and PolitiFact requires hours to days
- Existing automated tools provide verdicts without explaining the reasoning
- Many tools upload user-submitted articles to external servers, creating privacy risks

There is a clear need for an instant, explainable, privacy-preserving, and freely accessible fake news detection tool.

3. Objectives

- Develop an instant (sub-second) NLP-based fake news classification system
 - Provide full explainability — every verdict includes a linguistic justification
 - Ensure complete privacy — all processing done client-side with no data transmitted
 - Integrate Claude AI for optional expert-level deep contextual analysis
 - Build a professional, responsive, accessible web interface
 - Support session history, analytics, and report export functionality
 - Deploy as a zero-dependency single-file HTML application on GitHub Pages
-

4. Technologies Used

Layer	Technology	Purpose
Frontend Structure	HTML5	Semantic markup, ARIA accessibility, PWA manifest
Styling	CSS3	Custom properties, Grid/Flexbox, animations, glassmorphism
Logic	JavaScript ES6+	Analysis engine, DOM manipulation, API calls
AI Integration	Anthropic Claude API	Deep credibility analysis via claude-sonnet-4-6
Typography	Google Fonts	Playfair Display, Inter, JetBrains Mono
Client Storage	sessionStorage	Temporary history and analytics (tab-scoped)
Deployment	GitHub Pages	Free static hosting for single-file SPA

5. System Architecture

5.1 Overview

TruthScan is a Single-Page Application (SPA) with a layered architecture:

- UI Layer: Input Panel, Result Panel, History Strip, Analytics Cards
- Analysis Engine: NLP Heuristics Module, Claude AI Module (optional)
- Supporting Modules: HistoryManager, Analytics, ExportSystem, ThemeManager, ToastSystem

5.2 NLP Analysis Algorithm

The analysis uses a weighted heuristic scoring model:

```
Risk Score = (sensational_hits × 16) + (exclamations × 6) + (all_caps × 8) + (questions × 3)
Evid Score = (evidence_hits × 13) + (attribution × 11) + (numbers ? 15 : 0) + length_bonus
Verdict     = evidScore >= riskScore → "Credible" else "Disputed"
Confidence  = clamp(50 + |evidScore - riskScore| × 0.55, 42, 96)
```

5.3 Reading Level (Flesch-Kincaid)

```
FK = 206.835 - 1.015×(words/sentences) - 84.6×(syllables/words)
```

>80: Very Easy | >60: Standard | >40: Academic | ≤40: Technical

6. Features

6.1 Core Analysis

- Real-time NLP signal detection with debounced live preview
- Animated SVG radial confidence gauge (stroke-dashoffset animation)
- Risk cue detection: 29 sensational phrases, ALL-CAPS, excess punctuation
- Evidence cue detection: 26 factual patterns, 10 attribution phrases
- 4-category animated breakdown bars (Sensational, Punctuation, Specificity, Attribution)
- Flesch-Kincaid reading level assessment

6.2 AI Integration

- Optional Claude AI deep analysis (claude-sonnet-4-6)
- Expert 3-4 sentence credibility assessment with contextual reasoning
- Graceful fallback when API is unavailable
- Animated loading state with bouncing dots

6.3 UX & Productivity

- Session history (last 12 analyses) with click-to-reload
- Analytics dashboard: total, real, fake counts, avg confidence
- Copy formatted report to clipboard
- Export analysis as JSON with metadata
- Dark/Light mode toggle with smooth CSS custom property transition
- Keyboard shortcut: Ctrl+Enter to analyze
- Onboarding overlay for first-time users
- Skeleton loading state before results appear
- Toast notification system for all user actions
- Full responsive design: mobile, tablet, desktop
- ARIA labels, keyboard navigation, screen reader support

7. Security Implementation

7.1 XSS Prevention

All user-generated content is sanitized before DOM insertion via:

```
function escHtml(s){ return
String(s).replace(/&/g, '&amp;').replace(/</g, '&lt;').replace(/>/g, '&gt;
t;'); }
```

7.2 Input Validation

- Empty input check with user-friendly error toast
- Minimum 30 character requirement enforced before analysis
- Type checking on all variables before DOM operations

7.3 Data Privacy

- All NLP analysis runs locally — no text ever sent to a server
- Claude API is only called on explicit user action (clicking "Deep AI")

- sessionStorage used instead of localStorage (cleared on tab close)
 - No cookies, no analytics beacons, no third-party tracking
-

8. Future Enhancements

Phase 1 — Near Term (3-6 months)

- URL-based article fetching using fetch() and CORS proxy
- Multi-language support: Hindi, Telugu for regional news
- Browser extension for Chrome/Firefox (in-context analysis)
- PDF export using jsPDF or Puppeteer

Phase 2 — Medium Term (6-12 months)

- ML model integration: DistilBERT fine-tuned on FakeNewsNet dataset
- REST API backend using FastAPI/Node.js for batch processing
- User authentication and cloud-stored analysis history
- React Native mobile application

Phase 3 — Long Term (1-2 years)

- Real-time social media monitoring integration
 - Institutional dashboard for newsrooms and fact-checkers
 - Multi-model ensemble combining BERT, GPT, and Claude
 - Academic research tool with bias detection and source reliability scoring
-

9. Conclusion

TruthScan successfully demonstrates that a professional, AI-augmented fake news detection system can be built as a single-file browser application — requiring no server infrastructure, no installation, and no user accounts. The application achieves instant analysis (sub-1ms NLP), full explainability, complete privacy, and production-grade security, while delivering a polished, accessible user experience suitable for journalists, students, researchers, and the general public.

The project integrates cutting-edge AI (Claude API) with classical NLP heuristics and modern web engineering practices, resulting in a tool that is both academically rigorous and practically useful. The codebase follows security best practices including XSS prevention, input validation, and zero data retention by default.

TruthScan is ready for deployment on GitHub Pages and serves as a strong foundation for future enhancements including ML model integration, multi-language support, and a browser extension.

References

- Vosoughi, S., Roy, D., & Aral, S. (2018). The spread of true and false news online. *Science*, 359(6380), 1146-1151.
- Anthropic. (2025). Claude API Documentation. <https://docs.anthropic.com>
- Wang, W. Y. (2017). "Liar, Liar Pants on Fire": A New Benchmark Dataset for Fake News Detection. *ACL 2017*.
- Shu, K., et al. (2018). FakeNewsNet: A Data Repository with News Content, Social Context and Dynamic Information. *arXiv*.
- MDN Web Docs. (2025). Web APIs — sessionStorage, Clipboard API, ARIA. <https://developer.mozilla.org>
- Google Fonts. Playfair Display, Inter, JetBrains Mono. <https://fonts.google.com>